

Data Science Capstone - Literature Review

Project: MoodBand: An Inclusive Wearable System for Real-Time Mood Detection and Adaptive Sensory Feedback

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Literature Review

Affective Computing Fundamentals

Picard's foundational work [1] is where affective computing as a discipline really starts. Her central claim was that computers could be built to recognize and respond to emotional states, not just ignore them as noise. That idea sounds almost obvious now, but at the time it wasn't, and it's still the premise MoodBand rests on: that internal affective states leave measurable traces a system can learn to read.

Decades later, Conceição et al. [2] show what that premise looks like in practice with modern tools. They build on Russell's circumplex model, which maps emotion along two axes, valence and arousal, and use a deep-seeded clustering algorithm to classify emotional states from signals collected by a wearable wristband, testing it across three separate benchmark datasets. What makes this paper useful for MoodBand specifically is that it's working with the same kind of hardware form factor (a wrist-worn device) and the same underlying signal types we'd expect from a Fitbit or Apple Watch, and it still manages to map those signals onto discrete emotional categories reasonably well.

Wearable Physiological Sensing

Moving from theory to the specific question of wearables, Saganowski et al.'s systematic review [3] is a useful starting point. They find that heart rate, HRV, and electrodermal activity dominate the literature as the go-to signals for everyday emotion recognition, and they flag how much harder things get once you move out of the lab and into real, everyday conditions. That's exactly the setting MoodBand is targeting.

On the consumer-hardware side specifically, [4] is probably the single most reassuring paper on this list. Using the LifeSnaps dataset, built from actual Fitbit Sense smartwatches rather than research-grade sensors, the authors predict anxiety from everyday wearable data and get results suggesting a plain consumer device carries useful signal, not just noise.

That said, [5] is a good reality check on the same point. The authors ran into some genuinely annoying constraints when pulling Fitbit data for ML purposes: step counts reading zero while the device is just charging, HR data missing when the device isn't worn properly, all requiring manual filtering. The one that matters most for our design is that Fitbit only computes HRV while you're asleep and relatively still. That means we can't lean on HRV as a live, daytime signal at all.

Mood/Emotion Classification Models

Once you've got the signals, the question is what model actually does the classifying. The systematic review in [6] gives a pretty clear answer: classical machine learning methods, Random Forest and Support Vector Machines in particular, show up in the large majority of wearable-based stress detection studies, with Random Forest especially holding up well on EDA and HRV features.

[7] adds two things on top of that. First, the authors design their classification approach specifically for resource-constrained, on-device deployment. Second, they compare personalized models against generalized ones and find personalized models can win out even with limited data per person, which matters given our own user trial will be small by necessity.

Accessible Feedback Design

The last piece of MoodBand, getting feedback to the user through non-visual channels, is backed by a smaller but still useful set of papers. The iFeeling system [8] is about as close a precedent as exists: it takes a detected emotional state and renders it as a distinct vibration pattern on a phone, built explicitly with visually impaired users in mind.

[9] widens the lens a bit. It's a recent, large-scale systematic literature review spanning two decades (2004 to 2024) of haptic assistive tools for blind and low-vision users, drawing on 132 papers, and it's useful less for any single finding and more as a way to check iFeeling's design choices against the broader literature.

Finally, [10] grounds all of this in actual user experience. Yeo and Chew interviewed visually impaired people in Singapore about how they use mobile apps day to day, and the picture isn't entirely rosy: screen readers often don't work as reliably as you'd hope. Their recommendations explicitly call for combining audio and haptic feedback rather than relying on either alone, and for taking users' emotional wellbeing seriously as a design goal.

Synthesis

Put together, the pieces MoodBand depends on each have reasonable support behind them. Physiological signals can be mapped onto discrete moods [1], [2]. A consumer wearable like Fitbit is a workable, if imperfect, way to collect those signals [3]–[5]. Classical, lightweight models are a sensible fit for the classification step given the deployment constraints [6], [7]. And haptic feedback in particular has real precedent for communicating affective state to visually impaired users [8]–[10]. What doesn't exist yet, as far as this review found, is a single system that ties all four pieces together into one mobile-first pipeline.

References

- [1] R. W. Picard, *Affective Computing*. Cambridge, MA, USA: MIT Press, 1997.
- [2] M. A. Conceição, A. Dubois, S. Haustein, B. Miranda, and C. L. Azevedo, "Deep-seeded clustering for emotion recognition from wearable physiological sensors," *arXiv preprint arXiv:2308.09013*, 2023 (revised 2025).
- [3] S. Saganowski, B. Perz, A. G. Polak, and P. Kazienko, "Emotion recognition for everyday life using physiological signals from wearables: A systematic literature review," *IEEE Trans. Affective Comput.*, vol. 14, no. 3, pp. 1876–1897, 2023.
- [4] A. Velu, J. Ramesh, A. Ahmed, S. Ganguly, R. Aburukba, A. Sagahyroon, and F. Aloul, "Integrating Fitbit wearables and self-reported surveys for machine learning-based state–trait anxiety prediction," *Appl. Sci.*, vol. 15, no. 19, p. 10519, 2025.
- [5] Z. D. King, M. Khalid, H. Yu, K. Shibuya, K. Zanna, M. Majd, R. L. Brown, Y. Shen, G. Kypriotakis, T. Vaessen, C. P. Fagundes, and A. Sano, "Machine learning-based context aware EMAs: An offline feasibility study," *arXiv preprint arXiv:2506.15834*, 2025.
- [6] A. O. Pataca, E. Zdravevski, P. J. Coelho, N. M. Garcia, M. Deryuck, C. Albuquerque, and I. M. Pires, "Use of machine learning for predicting stress episodes based on wearable sensor data: A systematic review," *Comput. Biol. Med.*, vol. 198, pt. A, p. 111166, 2025.

[7] M. Irfan, A. Nawaz, A. K. Bulbul, R. Klén, A. Subasi, T. Westerlund, and W. Chen, "Emotion recognition with minimal wearable sensing: Multi-domain feature, hybrid feature selection, and personalized vs. generalized ensemble model analysis," in *Proc. 2025 IEEE Int. Conf. Bioinformatics and Biomedicine (BIBM)*, 2025, pp. 5447–5453.

[8] [Author names not independently confirmed — see note below] "iFeeling: Vibrotactile rendering of human emotions on mobile phones."

[9] C. Jiang, E. Kuang, and M. Fan, "How can haptic feedback assist people with blind and low vision (BLV): A systematic literature review," *arXiv preprint arXiv:2412.19105*, 2024.

[10] Y. P. H. Jesvin and C. K. Hao, "A framework for inclusive mobile application design: Insights from Singapore's visually impaired community," *ADADA*, vol. 29, suppl. 1, pp. 46–XX, 2024.